1.33 Differentiation: Graphs of Gradient Functions

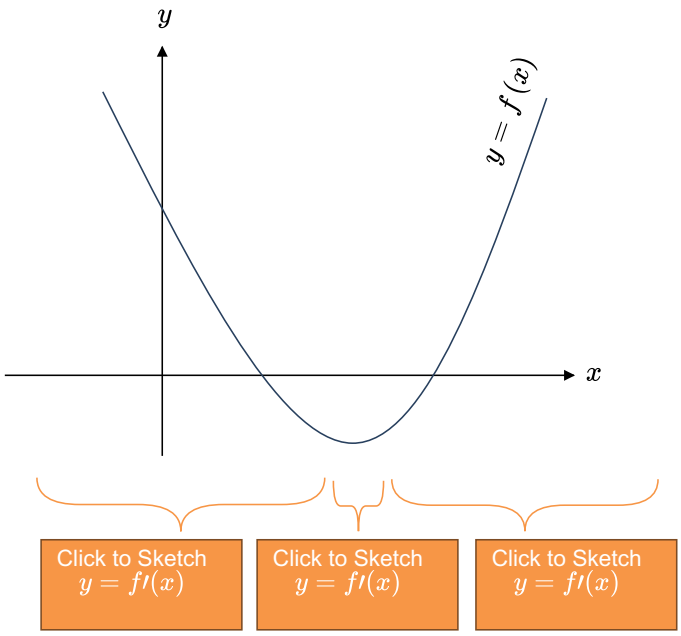
Chapter 12 · Original slide 64

Sketching Gradient Functions

64

The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs. But in other cases **you won't be given the function explicitly**, but **just the sketch**.

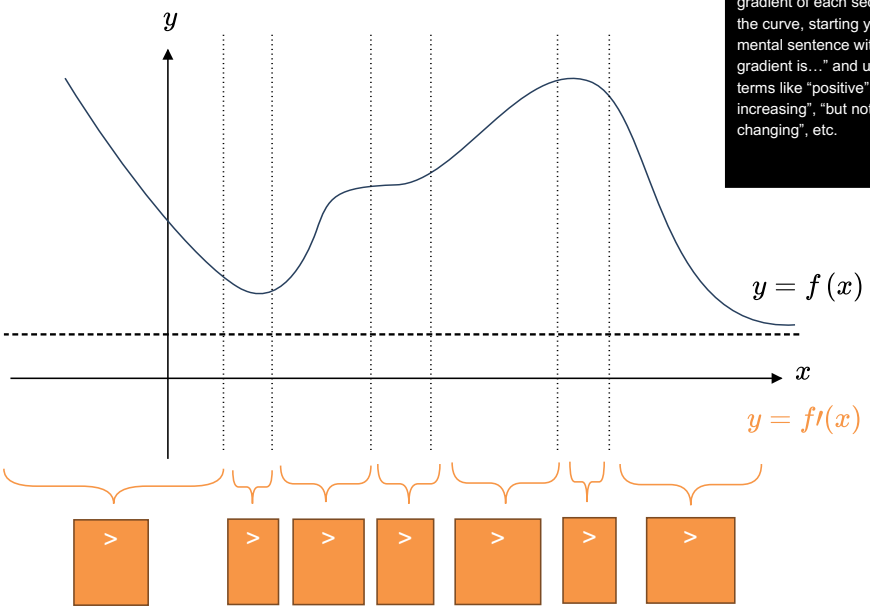


Chapter 12 · Original slide 65

A Harder One

65

Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with "The gradient is…" and using terms like "positive", "but increasing", "but not changing", etc.

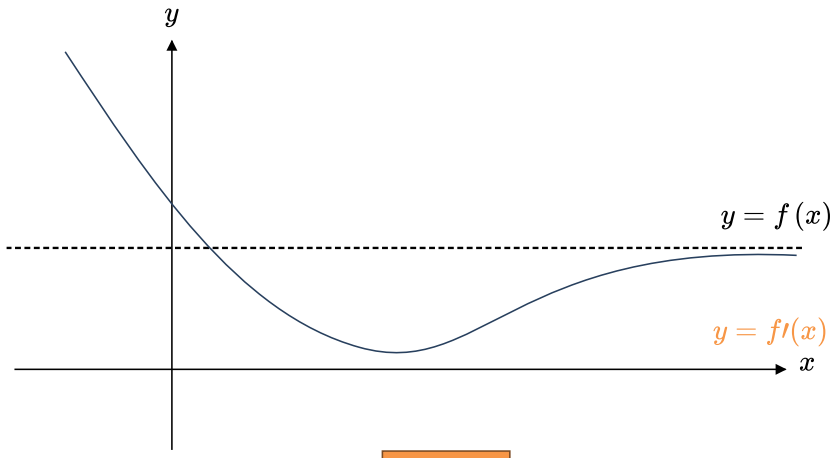


1.33 Differentiation: Graphs of Gradient Functions

Chapter 12 · Original slide 66

Test Your Understanding

66



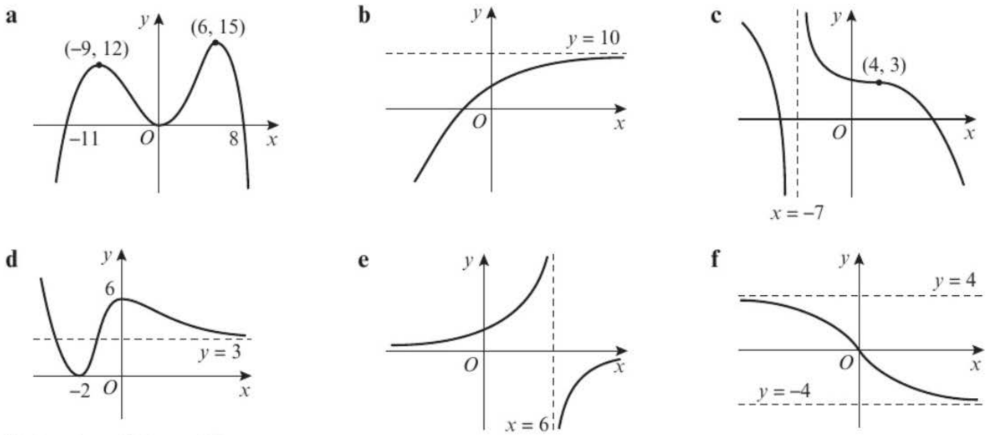
Solution >

Chapter 12 · Original slide 67

Exercise 12J – Pearson Pure Y1 page 278

67

1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.



Solutions here

1.33 Differentiation: Graphs of Gradient Functions

Chapter 12 · Original slide 68

Exercise 12J – Pearson Pure Y1 page 278

68

2\quad $\mathrm{f}(x) = (x+1)(x-4)^2$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On a separate set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Show that $\mathrm{f}'(x) = (x-4)(3x-2)$.
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

Chapter 12 · Original slide 69

Extension

69

1 [MAT 2015 1B]

$$f(x) = (x+a)^n$$

where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:
A) always be odd
B) always be even
C) depend on a but not n
D) depend on n but not a
E) depend on both a and n.
F)